NetraJyoti AI

Technical Product Requirements Document (Technical PRD)

A Bengali-first, safety-guidance MVP for people seeking appropriate eye-care services in rural West Bengal. The product provides safe next-step guidance and care navigation; it does not diagnose eye disease.

Document intent. This technical PRD translates the capstone PRD, approved user journey, and current MVP implementation into build-ready requirements for engineering, quality assurance, operations, and review.

Document control	Value
Product	NetraJyoti AI MVP
Version	1.0
Primary language	Bengali-first interface with smaller English support text
Client	React + TypeScript + Vite + Tailwind CSS / PWA-ready web client
Server	Java 21 + Spring Boot 3 REST API
Data	PostgreSQL with Spring Data JPA / Hibernate
Safety posture	Deterministic reviewed rules decide urgency; AI never decides urgency or diagnoses
Status	MVP technical baseline

1. Product and technical context

NetraJyoti AI addresses the delay, fear, and confusion experienced by Bengali-speaking users who are unsure whether an eye concern needs urgent attention, where to seek care, and which advice to trust. The application turns a short guided interaction into a safe, explainable next step: urgent eye-care advice, routine vision/eye-check guidance, or human-support guidance.

The system is intentionally an information and care-navigation product. It must never present a disease label, claim clinical diagnosis, or allow generative AI to override reviewed safety rules.

1.1 Product objectives

- Give Bengali-speaking users a low-literacy-friendly, mobile-first way to describe fixed eye symptoms.
- Identify urgent warning signs through deterministic, reviewed rules and immediately route to urgent care guidance.

- Offer practical service navigation: verified-care discovery, district/town selection, PHC referral, eye clinic/vision centre direction, family sharing, and human support.
- Create a concise caregiver message containing selected symptoms, approved next step, and optional consented patient information.
- Use controlled AI only to improve the routine caregiver summary after deterministic routing and backend safety validation.

1.2 Scope and non-goals

In MVP scope	Out of scope / prohibited
Bengali-first symptom selection, optional voice/text capture, reviewed safety route, service direction, caregiver message and share/copy.	Disease diagnosis, medication recommendation, treatment plan, triage model training, or autonomous clinical decision-making.
Demo district/town or block service selection and curated demo facility data.	Claims that availability, hours, or facility details are live unless an approved directory integration is introduced.
Optional OpenAI-generated caregiver summary behind a feature flag and server validation.	Sending raw free-text, recording audio, or patient details to AI without an approved privacy and consent design.
PWA-capable browser experience.	Native mobile application store distribution in MVP.

2. End-to-end workflow and routing

The application must preserve the following deterministic workflow. The UI wireframe labels each screen with its entry trigger to make the route auditable during product review and testing.

Stage	User/system action	Transition rule
1. Consent	User reads value and safety notice, then selects consent.	Consent enables “Start safely”. No patient details are requested.
2. Symptoms	User selects one or more fixed symptom options; voice/text is optional context.	Submitted selections move to warning-sign review. Optional detail never sets urgency.
3. Urgent warning check	User answers urgent-warning questions.	Reviewed rules prepare one of three safe routes.
4. Safe guidance check	UI confirms data received and warning signs checked.	Urgent → 5A; routine → 5B; unclear/human support → 5C.
5A urgent	Immediate care advice; navigation, family sharing, or care confirmation.	Sub-paths 5A.1–5A.6 retain access to the urgent result.
5B routine	User chooses district and town/block.	Approved routine guidance opens Step 6.
6–8 routine	Recommendation → service direction → caregiver summary.	Patient details are optional and only used after explicit share consent.
5C human support	User is guided to ASHA/PHC/trusted-family options.	Location/area can be used only to find nearby support.

2.1 Deterministic routing contract

Safety invariant. The backend rules module is the sole authority for urgency. UI state, optional voice/text, AI output, patient profile fields, and service selection must not alter the urgency decision.

Route	Entry condition	Required guidance and actions
Urgent (5A)	At least one reviewed urgent warning sign is selected.	Show “seek eye care without delay”, selected concerns, verified-care discovery, PHC support, family share, copy, care confirmation, and restart.
Routine (5B–8)	No urgent warning sign; issue can follow approved routine route.	Offer district/town selection, visit recommendation, service direction, listening support, caregiver summary, sharing, and restart.
Human support (5C–5C.2)	Unclear concern, “other” response, or user requests human support.	Explain that a local health worker, PHC, eye clinic, or trusted family member can help; offer area-based support discovery and sharing.

2.2 Urgent-route state transitions

Current stage	Trigger	Next stage
5A	Find verified care	5A.1
5A.1	Use current location	5A.2
5A.1	District/block or PHC support	5A.3
5A.2	Location allowed	5A.4
5A.2	Permission declined / district option	5A.3
5A.3	District + block selected; view services	5A.4
5A	Share urgent message	5A.5
5A	Continue to care	5A.6
5A.4/5A.5/5A.6	Return urgent result	5A

3. Solution architecture

NetraJyoti is designed as a browser-based, API-driven application. The client retains transient interaction state while the server owns reviewed routing, controlled-AI validation, service-direction data APIs, and audit-ready telemetry. PostgreSQL is the system database for configuration and operational data; personally identifying details should not be persisted in the MVP unless a separate consent and retention design is approved.

Layer	Technology	Responsibilities
Presentation	React, TypeScript, Vite, Tailwind CSS	Bengali-first responsive screens, accessible controls, workflow state, Web Share/Clipboard integration, optional browser speech input, PWA manifest and service worker.
API	Java 21, Spring Boot 3, REST, Spring Security, Bean Validation	Route evaluation, service lookup, AI-summary endpoint, feature flags, validation, rate limits, error contracts, audit events.
Safety domain	Deterministic Java rules module	Maps fixed symptom and urgent-sign identifiers to urgent/routine/human result. Versioned and independently unit-tested.
Persistence	PostgreSQL, Spring Data JPA/Hibernate	Facilities, locations, reviewed content, rules/version metadata, feature-flag defaults, non-identifying analytics/audit records.
AI integration	OpenAI REST API called only from backend	Optional Bengali caregiver summary only after a safe route exists; validates response against schema and falls back to approved Bengali content.
Delivery	Vercel/Netlify client; Render/Railway/AWS/Azure API + PostgreSQL	TLS, environment-specific configuration, logs, backups, health checks, and deployment controls.

3.1 Context data flow

- The browser sends only fixed symptom identifiers and warning-sign answers to the route API. The API returns a reviewed route code and approved content key.
- The client uses district and town/block only to request a service-direction result. Browser geolocation is requested only after the user selects the location action.
- The caregiver-summary request is made only for a routine route, only when the “Enable AI summary” flag is true, and only through the Java backend. The OpenAI API key is never sent to the client.
- If AI is off, unavailable, times out, fails validation, or produces disallowed text, the backend returns the approved Bengali fallback. The user flow remains complete.

4. Frontend requirements

Area	Technical requirement
State model	Use a typed workflow state machine or reducer. Persist only low-risk, session-scoped UI state if needed; reset on “Start again”.
Internationalisation	Bengali is the primary label and English is secondary, visually smaller text. Avoid untranslated utility labels on user-facing screens.
Accessibility	Large touch targets (minimum 44px), high contrast, clearly visible selection states, keyboard focus states, semantic buttons/labels, status announcements, and readable bilingual hierarchy.

Area	Technical requirement
Progression	Every stage must expose an understandable primary action and a safe back/return/restart route where appropriate. The wireframe “Entered from / Trigger” traceability is documentation, not an end-user control.
Voice	Use browser speech recognition only as optional input convenience. Display captured text for confirmation. Failure/unavailability must leave typed/fixed-option flow fully usable.
Sharing	Use Web Share API where supported; otherwise offer Copy. Provide clear success/error status. Never infer that sharing completed successfully.
PWA	Installable manifest, responsive layout, HTTPS-only deployment, cache only static application assets; do not cache API responses containing patient data.

5. Backend services and API contract

REST endpoints must use JSON, UTF-8, API versioning under /api/v1, validation errors with machine-readable codes, and a consistent error envelope. Endpoint names below are the target MVP contract; implementation may add internal fields without breaking documented client behavior.

Method and path	Purpose	Core request / response
POST /api/v1/routing/evaluate	Evaluate deterministic reviewed safety route.	Request: symptomIds, urgentSignIds. Response: route (URGENT|ROUTINE|HUMAN_SUPPORT), ruleVersion, contentKey, reasons[]; no diagnosis.
GET /api/v1/locations/districts	List supported demo districts.	Response: bilingual district id/name values.
GET /api/v1/locations/districts/{id}/areas	List towns, cities, or blocks for a district.	Response: bilingual area id/name values.
GET /api/v1/services	Return approved care/support options.	Query: districtId, areaId, route, optional latitude/longitude. Response: service cards, contact, verifiedAt, demo flag.
POST /api/v1/summaries/caregiver	Return routine caregiver message.	Request: safe route context, selected symptoms, optional consented patient detail. Response: approved or AI-generated summary, source, fallbackUsed.
GET /api/v1/features	Read client-safe feature flags.	Response: enableAiSummary and other non-secret presentation flags.
GET /actuator/health	Platform liveness/readiness.	No sensitive operational details exposed publicly.

5.1 Routing request and response rules

API validation. Reject missing or unknown fixed identifiers; never route based on arbitrary free text. Return HTTP 400 with a stable error code for invalid input. Return HTTP 429 for rate limits and a safe generic fallback message for unexpected failure.

Field	Rules
symptomIds	Array of approved catalogue IDs. At least one is required for normal progression.
urgentSignIds	Array of approved warning-sign IDs. The Java rules module evaluates only these identifiers.
route	One of URGENT, ROUTINE, HUMAN_SUPPORT. Must be accompanied by ruleVersion and approved content key.
districtId / areaId	Optional until service-direction stage; must be known catalogue IDs when supplied.
patient details	Optional name, age, address, phone. Used only in a share-preview request after explicit consent. Do not log values or persist by default.
ai summary input	Routine route only. Must exclude raw audio and must be schema-limited before an external

Field	Rules
	call.

6. Data model

Entity	Key fields	Notes
SafetyRuleSet	id, version, activeFrom, activeTo, approvedBy, status	Immutable or versioned record. Exactly one active approved rule set per environment.
SymptomCatalog	id, bengaliLabel, englishLabel, category, active	Fixed user-selectable options; labels owned by reviewed content process.
UrgentSignCatalog	id, bengaliLabel, englishLabel, active	Fixed answers consumed by deterministic routing.
ServiceLocation	id, type, districtId, areaId, name, address, phone, verifiedAt, active, demo	Vision centre, eye clinic, hospital, PHC, ASHA contact/support directory.
ApprovedContent	key, locale, route, body, version, approvedBy, active	Fallback messages and safety copy. Store Bengali and English separately.
FeatureFlag	key, enabled, environment, updatedAt, updatedBy	Includes ENABLE_AI_SUMMARY. Default false in production until approval.
AuditEvent	eventType, route, ruleVersion, contentVersion, timestamp, correlationId	No patient identifiers or raw free-text. Supports safety traceability.
SummaryRequestAudit	source, fallbackUsed, validationStatus, latencyMs, timestamp	No sensitive prompt or response text by default; keep technical metadata only.

7. Controlled AI / GenAI design

Controlled AI is an enhancement, not a routing mechanism. Its only MVP purpose is to produce a short, supportive Bengali caregiver summary after a routine route has already been determined by reviewed rules. The application must remain useful when the feature flag is off.

7.1 Feature flag and decision flow

Step	Required behavior
1. Route first	Call deterministic routing endpoint and accept only ROUTINE route for AI summary eligibility.
2. Check flag	Read ENABLE_AI_SUMMARY server-side. If false, return approved Bengali fallback immediately.
3. Minimise input	Send only allowlisted context: route, selected catalogue labels, approved next step, and optionally consented share fields. Never include raw audio.
4. Generate	Backend invokes OpenAI with a constrained system instruction and structured output request. API key stays in server environment variables.
5. Validate	Check language, max length, schema, prohibited diagnosis/treatment claims, unchanged safety instruction, and no invented facility details.
6. Fallback	If timeout, API error, malformed output, safety rejection, or rate limit occurs, return approved content with fallbackUsed=true.
7. Present	Client labels the result as a family/caregiver summary and keeps share/copy optional. It must not imply medical diagnosis or replace care guidance.

7.2 AI guardrails

- Prompt instruction: do not diagnose, name a disease, prescribe medicine, change urgency, or claim a clinician has reviewed the user.
- Use structured JSON output with fields such as bengaliSummary, englishSummary, safetyNoticeRetained, and validationStatus; reject any unexpected fields.
- Post-generation policy validation must reject prohibited diagnostic terms and medical instructions, enforce approved action phrases, and cap output length.
- Apply request timeout, retry only safe idempotent calls, per-user/IP rate limiting, and circuit-breaker behavior. Fallback content must be local and immediately available.
- Log only metadata needed for reliability and review. Do not log OPENAI_API_KEY, raw patient data, full prompts, or full model output by default.

Feature flag. ENABLE_AI_SUMMARY is configured through environment/application configuration and exposed to the client only as a boolean capability. Turning it off must need no client redeploy and must force the approved fallback.

8. Security, privacy, and safety controls

Control area	MVP requirement
Secrets	OPENAI_API_KEY, database credentials, and CORS origins live in environment variables/secret

Control area	MVP requirement
	manager only. Never commit .env files or expose keys in JavaScript bundles.
Transport	HTTPS in every deployed environment; secure headers; strict CORS allowlist for known frontend origins.
Authentication	Public user routes can be anonymous with rate limiting. Administrative updates (service data, rules, content, flags) require authenticated, role-based access.
Patient details	Keep client-side until consented sharing. Explain local-use purpose. Do not persist or analytics-track name, age, address, phone, audio, or raw free-text in MVP.
Location	Request browser location only after explicit action. Provide district/town/block alternative. Do not silently retain coordinates.
Content governance	Rules, urgent copy, service entries, and fallback messages require version, approval owner, and reviewed-at metadata.
Auditability	Store route code, rule/content version, feature-flag state, and technical outcome without user identifiers.

9. Non-functional requirements

Category	Target / requirement
Availability	Core deterministic routing and approved fallback operate without dependency on OpenAI. Degraded mode must keep all critical guidance available.
Performance	p95 routing API < 500 ms excluding network; service lookup p95 < 750 ms; AI summary has explicit timeout and non-blocking fallback.
Reliability	No state transition may lose selected symptoms before a result; all critical screens offer a path back, return to result, or restart.
Mobile usability	Functional from 320 px width upward; primary actions reachable with thumb-friendly controls; no hover-only interaction.
Accessibility	WCAG 2.1 AA intent: colour contrast, text resize, focus handling, labels, error messaging, and touch target sizing.
Observability	Correlation IDs, structured logs, health checks, route distribution, rule version usage, facility lookup errors, AI fallback rate, and API latency dashboards.
Maintainability	Typed contracts; centrally defined labels/content keys; linting; unit, integration, and end-to-end tests; conventional environment configuration.
Data retention	No persistent patient data in MVP. Define explicit retention and deletion policy before any future collection.

10. Test strategy and acceptance criteria

Test level	Coverage
Java unit tests: JUnit 5 + Mockito	Every rule combination; precedence of urgent signs; invalid IDs; content-key selection; feature flag branching; AI validator and fallback.
Spring integration tests	REST validation/error envelope, PostgreSQL repositories, authorised administration, CORS configuration, service filtering, and actuator health.
React Testing Library	Consent enablement, option selection, bilingual labels, back/restart behavior, screen transitions, sharing status, and fallback presentation.
Playwright journeys	Urgent full path 5A–5A.6; routine 5B–6–7–8; human 5C–5C.2; no-location alternative; AI on/off/error behavior; mobile viewport.
Manual content review	Bengali naturalness, English secondary readability, no overlapping UI, content governance, service-data disclaimer, and non-diagnosis boundary.
Security checks	Secret scan, dependency scan, CORS test, rate-limit behavior, no PII in logs, and OWASP-relevant API tests.

10.1 Release acceptance criteria

- Every fixed symptom/warning-sign test maps to a reviewed expected route, with no AI dependency.

- An urgent result can reach service discovery, location permission, district/block selection, facilities, sharing, care confirmation, and restart without a dead end.
- A routine result can reach district/town selection, visit guidance, service direction, caregiver summary preview, share/copy, and restart.
- A human-support result can reach ASHA/PHC/family support and area-based nearby-support options.
- With ENABLE_AI_SUMMARY=false, the Step 8 approved Bengali fallback is shown. With it true, unsafe/unavailable AI output still returns the same safe fallback.
- No view claims diagnosis, treatment recommendation, or verified real-time service availability unless supported by approved data.
- The deployed frontend and backend run over HTTPS, no secret is included in the client build, and the health endpoint is monitored.

11. Deployment and operating model

Environment	Deployment requirements
Local development	Docker Compose may run PostgreSQL. Frontend uses Vite dev server; Spring Boot runs with local profile; .env is ignored by Git and copied from .env.example.
Test / staging	Separate database, non-production secrets, test service data, controlled feature flags, and end-to-end smoke test on every release candidate.
Production	Static frontend on Vercel/Netlify; Spring Boot and PostgreSQL on managed platform; TLS, backups, database migrations, monitoring, restricted admin access, and deployment rollback.
Configuration	CORS_ORIGIN, database URL/user/password, OPENAI_API_KEY, ENABLE_AI_SUMMARY, service directory mode, rate limits, and log level are environment-specific.
Database migration	Use Flyway or Liquibase before production launch. Migrations must be reversible where possible and version safety/content records.

12. Delivery plan and ownership

Workstream	Deliverable	Primary owner
Safety and content	Approved symptom catalogue, urgent signs, rules, Bengali/English copy, version governance.	Product + clinical/content reviewer
Frontend	Responsive bilingual flow, PWA, share/copy, voice fallback, accessibility and UI tests.	Frontend engineer
Backend	Routing rules API, service directory API, flags, AI guardrails, observability.	Java backend engineer
Data and operations	PostgreSQL schema, migrations, service-data update process, backups and access controls.	Backend / platform owner
Quality assurance	Test matrix, route regression suite, device checks, error/fallback validation.	QA engineer
Outreach operations	Eye-camp, vision-centre, PHC, clinic/hospital information verification	Outreach coordinator

Workstream	Deliverable	Primary owner
	and update cadence.	

Appendix A. Technical decision summary

The MVP chooses a deterministic Java rules module because the product's most consequential decision is urgency. React/PWA maximises Android-friendly access without an app-store barrier. PostgreSQL and JPA provide governed content and service data. OpenAI is intentionally isolated behind a Spring Boot service, feature flag, schema validation, safety policy checks, and an approved fallback. This architecture keeps the core patient value available even when optional AI or location services are unavailable.

Future AI and Agentic Capability Requirements (Post-MVP / Controlled Pilot)

This is a future-state option and is not part of the current MVP safety-routing path. It may proceed only after clinical/content approval, privacy impact review, Bengali usability validation, partner data governance and controlled-pilot acceptance.

Safety invariant: reviewed deterministic Java rules remain the only component that selects URGENT, ROUTINE or HUMAN_SUPPORT. A model, agent, MCP connector, skill, voice input, free text, location or patient detail must never change that outcome.

Future component	Purpose	Requirement and guardrail
AI gateway / model client	Generate a plain-Bengali explanation or caregiver summary from a fixed approved result.	Backend-only calls; versioned model configuration; allowlisted input; strict timeout, cost cap, schema/policy validation and Bengali fallback.
Agent orchestrator	Execute a bounded workflow after routing, for example approved content -> summary -> validation -> review.	No autonomous planning outside a defined workflow; no direct user contact; no route-changing tool; every action correlation-ID audited.
Skill registry	Expose small, typed skills such as summary formatting, translation, verified service lookup and data-quality flagging.	Skills have explicit input/output contracts, least privilege, tests, owner and version; no skill can diagnose, prescribe or publish changes.
MCP gateway / connectors	Connect read-only to approved service-directory or knowledge-content systems.	Allowlisted servers/tools only; authenticated server-side access; read-only by default; source and verified-at metadata returned to user.
Safety validator and fallback	Reject outputs that conflict with approved route or introduce unsafe claims.	Block diagnosis, medicines, invented facilities, emergency reassurance, route overrides and unsupported medical claims; return approved fallback on failure.
Human review queue	Review coordinator drafts, data-quality flags and future content changes.	No autonomous publish, booking, messaging or service-data correction; auditable human approval is mandatory.

Reference workflow: deterministic route -> approved result content -> ENABLE_AI_SUMMARY feature flag -> bounded agent and skills -> model and optional approved MCP retrieval -> safety validator -> user-reviewable output OR approved Bengali fallback.

Non-functional control	Future-state requirement
Privacy	Do not send raw voice, unapproved free text, identifiers, phone, address or precise location to the model. Use consented, minimal, allowlisted fields only.
Observability	Log non-identifying correlation ID, route, rule/content version, flag state, connector/source identifier, latency, validation result and fallback status.
Resilience	The system remains fully safe and usable when the model, MCP server or any skill fails, times out or is disabled.
Change governance	Each prompt, skill, connector and model configuration has a named owner, version, evaluation evidence and rollback path.